Plan

- Subspaces Associated with a Matrix
- Rank Nullity Theorem
- The Fundamental Theorem of Invertible Matrices

Subspaces Associated with a Matrix

Let A be an $m \times n$ matrix with entries in $\mathbb{F}$ ($\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$).

1. The null space of A: $\text{null}(A) = \{\mathbf{x} \in \mathbb{F}^n \mid A\mathbf{x} = \mathbf{0}\}$.
   - A subspace of $\mathbb{F}^n$.
   - The solution space of $A\mathbf{x} = \mathbf{0}$.
2. The column space of A: $\text{col}(A) = \{A\mathbf{x} \mid \mathbf{x} \in \mathbb{F}^n\}$.
   - The subspace of $\mathbb{F}^m$ spanned by the columns of A.
   - The set of vectors $\mathbf{b} \in \mathbb{F}^m$ for which $A\mathbf{x} = \mathbf{b}$ has a solution.
   - $A\mathbf{x} = \mathbf{b}$ has a solution $\iff$ $\mathbf{b} \in \text{col}(A)$.
3. The row space of A: $\text{row}(A) := \{\mathbf{x}^t A : \mathbf{x} \in \mathbb{F}^m\}$.
   - The set of all linear combinations of the rows of A.
   - A subspace of $\mathbb{F}^n$, considering $\mathbb{F}^n$ as set of row vectors.
   - $\mathbf{u} \in \text{row}(A) \iff \mathbf{u}^t \in \text{col}(A^T)$

Take $A = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 2 & 4 & 2 & 2 \\ 3 & 6 & 4 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

- Are A and B related?
- What is row(A)? row(B)? Are they same?
- What is col(A)? col(B)? Are they same? Are their dimensions same?

Result

- *Let A and B be $m \times n$ matrices. Then A and B are row equivalent iff $B = PA$ for some invertible $m \times m$ matrix P.*
- *Let the matrices B and A be row equivalent. Then $row(B) = row(A)$.*
- *For any matrix A, $row(A) = row(RREF(A))$.*
- *For any A, the non-zero rows of $RREF(A)$ form a basis of $row(A)$.*

Result

- *Let P be an $m \times m$ invertible matrix. Then $\mathbf{a}_1, \ldots, \mathbf{a}_k \in \mathbb{F}^m$ are linearly independent iff $P\mathbf{a}_1, \ldots, P\mathbf{a}_k \in \mathbb{F}^m$ are linearly independent.*

$$\boxed{\begin{aligned} \alpha_1 P\mathbf{a}_1 + \cdots + \alpha_k P\mathbf{a}_k = 0 \quad &\Longleftrightarrow \quad P(\alpha_1\mathbf{a}_1 + \cdots + \alpha_k\mathbf{a}_k) = 0 \\ &\Longleftrightarrow \quad \alpha_1\mathbf{a}_1 + \cdots + \alpha_k\mathbf{a}_k = 0 \end{aligned}}$$

- *Suppose $A = [\mathbf{a}_1\ \mathbf{a}_2\ \cdots\ \mathbf{a}_n]$ and $RREF(A) = [\mathbf{r}_1\ \mathbf{r}_2\ \cdots\ \mathbf{r}_n]$ has k nonzero rows. If $\mathbf{r}_{j_1} = \mathbf{e}_1, \ldots, \mathbf{r}_{j_k} = \mathbf{e}_k$, then*
  - $\mathbf{e}_1, \ldots, \mathbf{e}_k$ *is a basis of $col(RREF(A))$;*
  - $\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_k}$ *is a basis of $col(A)$.*

Result

For any matrix A, $\dim col(A) = \dim row(A) = rank(A)$.

- Column rank of A := $\dim \mathrm{col}(A)$;
- Row rank of A := $\dim \mathrm{row}(A)$;
- Thus, column rank of A = row rank of A = $\mathrm{rank}(A)$.
- Moreover, $\mathrm{rank}(A^t)$ = column rank of A^t = row rank of $A = \mathrm{rank}(A)$, i.e., $\mathrm{rank}(A^t) = \mathrm{rank}(A)$.

Nullity: The dimension of $\mathrm{null}(A)$ is called the nullity of A, and is denoted by $\mathrm{nullity}(A)$.

- $\mathrm{nullity}(A) =$ the no. of free variables in the system $A\mathbf{x} = \mathbf{0}$.

Result (**Rank Nullity Theorem**)

Let A be an $m \times n$ matrix. Then

$$rank(A) + nullity(A) = n.$$

Finding bases for $\mathrm{row}(A), \mathrm{col}(A)$ and $\mathrm{null}(A)$

Let A be a given matrix and let R be the reduced row echelon form of A.

1. For $\mathrm{row}(A)$: The nonzero rows of R form a basis for $\mathrm{row}(A)$.

2. For $\mathrm{null}(A)$: Solve the leading variables of $R\mathbf{x} = \mathbf{0}$ in terms of the free variables, set the free variables equal to parameters, substitute back into $\mathbf{x}$, write the result as a linear combination of k vectors (where k is the number of free variables). These k vectors form a basis for $\mathrm{null}(A)$.

3. For $\mathrm{col}(A)$: The columns of A that correspond to the columns of R containing the leading 1's form a basis for $\mathrm{col}(A)$.

   Alternately, find a basis for $\mathrm{row}(A^t)$. The transposes of these vecors will be a basis for $\mathrm{col}(A)$.

Example

Find bases for (i) the row space, (ii) the column space and (iii) the null space of the following matrix:

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 2 & 4 \\ 4 & 6 & 2 \end{bmatrix}, \quad RREF(A) = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix}.$$

(i) $\{[1, 0, 5], [0, 1, -3]\}$

(ii) $\left\{ \begin{bmatrix} -5 \\ 3 \\ 1 \end{bmatrix} \right\}$

(iii) $\left\{ \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 6 \end{bmatrix} \right\}$

Result (Fundamental Theorem of Invertible Matrices: II)

Let A be an $n \times n$ matrix. Then the following statements are equivalent.

1. A *is invertible.*
2. A^t *is invertible.*
3. $A\mathbf{x} = \mathbf{b}$ *has a solution for every* $\mathbf{b}$ *in* $\mathbb{F}^n$.
4. $A\mathbf{x} = \mathbf{b}$ *has a unique solution for every* $\mathbf{b}$ *in* $\mathbb{F}^n$.
5. $A\mathbf{x} = \mathbf{0}$ *has only the trivial solution.*
6. *The reduced row echelon form of* A *is* I_n.
7. *The rows of* A *are linearly independent.*
8. *The columns of* A *are linearly independent.*
9. $\mathrm{rank}(A) = n$.
10. A *is a product of elementary matrices.*

11. nullity$(A) = 0$.

12. col$(A) = \mathbb{F}^n$.

13. The column rank of A is n.

14. row$(A) = \mathbb{F}^n$.

15. The row rank of A is n.

Exercise

Show that the vectors $[1, 2, 3]^t, [-1, 0, 1]^t$ *and* $[4, 9, 7]^t$ *form a basis for* $\mathbb{C}^3$.

You can solve the problem in several different ways! Try.

Result

Let A be an $m \times n$ matrix. Then

1. $\mathit{rank}(A^*A) = \mathit{rank}(A)$.
   (Here, A^* is the conjugate transpose of A.)
2. *The $n \times n$ matrix A^*A is invertible if and only if $\mathit{rank}(A) = n$.*

Proof. (1) Claim: A^*A and A have same null space:

First, $A\mathbf{x} = \mathbf{0} \implies A^*A\mathbf{x} = \mathbf{0}$, i.e., $\mathrm{null}(A) \subseteq \mathrm{null}(A^*A)$.

Suppose $A^*A\mathbf{x} = \mathbf{0}$ and $A\mathbf{x} = [y_1, \ldots, y_m]^t$. Then

$$0 = \mathbf{x}^*A^*A\mathbf{x} = (A\mathbf{x})^*(A\mathbf{x}) = |y_1|^2 + \cdots + |y_m|^2,$$ and therefore

$A\mathbf{x} = [y_1, \ldots, y_n]^t = \mathbf{0}$. Thus, $\mathrm{null}(A) = \mathrm{null}(A^*A)$.

Now, use Rank-Nullity-Theorem.

(2) A^*A is invertible iff $\mathrm{rank}(A^*A) = n$ iff $\mathrm{rank}(A) = n$.

Result

- *If T, S are invertible, and TA and AS are defined, then*

$$\mathit{rank}(TA) = \mathit{rank}(A) = \mathit{rank}(AS).$$

[**Hint.** *TA and A are row equivalent, $(AS)^t$ and A^t are row equivalent.*]

- *If AB is defined then*

$$\mathit{rank}(AB) \leq \min\{\mathit{rank}(A), \mathit{rank}(B)\}.$$

[**Hint.** *$\mathit{row}(AB) \subseteq \mathit{row}(B)$ and $\mathit{col}(AB) \subseteq \mathit{col}(A)$.*]

- *If $A + B$ is defined then*

$$\mathit{rank}(A + B) \leq \mathit{rank}(A) + \mathit{rank}(B).$$

[**Hint.** *Rows of $A + B$ are linear combinations of the union of rows of A and B.*]

Result (Rank Factorization)

Let A be an $n \times n$ matrix of rank r, where $1 \leq r < n$. Then there exist elementary matrices $E_1, \ldots, E_p$ and $F_1, \ldots, F_q$ such that

$$E_1 \cdots E_p A F_1 \cdots F_q = \begin{bmatrix} I_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}.$$

Hint. There are $E_1, \ldots, E_p$ such that

$E_1, \ldots, E_p A = R = RREF(A)$.

$$R = [\cdots \mathbf{e}_1 \cdots \mathbf{e}_2 \cdots \mathbf{e}_r \cdots], \text{ and } R^t = \begin{bmatrix} \vdots \\ \mathbf{e}_1^t \\ \vdots \\ \mathbf{e}_2^t \\ \vdots \\ \mathbf{e}_r^t \\ \vdots \end{bmatrix}.$$

There are $F_1, \ldots, F_q$ such that

$$F_q, \cdots, F_1 R^t = \begin{bmatrix} I_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}.$$

Then $E_1 \ldots E_p A F_1^t \cdots F_q^t = \begin{bmatrix} I_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix}$.